# FALCON 4K

### 3.1.1-OA Changelog:

**- New MESSAGES page in History (replaces the RX/TX log):** messages received by FMessage are saved automatically in external flash.

**- New F1/F2/M long shortcut MESSAGES:** opens History directly on the MESSAGES page.

**- New F1/F2/M long shortcut FMESSAGE:** opens the FMessage app (double beep if not installed).

**- New REVERSE function:** REVERSE: line in the channel editor, REVERSE FQ shortcut, “R” indicator on the main screen.

**- New SetRxG menu:** RX noise gate ON/OFF, default ON.

**- General Menu categorized.**

**- New FCatch history:** apps can now save a block of their own data, so FCatch (Close Call capture) keeps its 17-entry history between launches. Requires Falcon 3.1.1.

Small other fixes:
- Received audio aligned on F4HWN (fixes USB audio after FM).
- Compander fixed for the BK4829 (no longer overwrites the noise gate and audio filter).
- TX powers aligned on F4HWN (MID and HIGH).
- Fixed scan sensitivity on weak dBm signals.
- Fixed Full Watch VFO channel bubble display.
- 1750 Hz tone now heard in the speaker while transmitting.
- VFO frequency and channel saved 3 s after the last change (less flash wear), immediately before TX, menu, scans and apps.
- Held shortcut keys (F1, F2, M long) no longer trigger extra actions.
- History keys popup no longer flickers under the scan status bar.
- Channel editor popups renamed EDIT/NEW/DELETE CHANNEL, LIST and BAND.
- Persistent popups restyled: rounded corners, right/bottom shadow, rounded white margin.
- Rounded highlight in menu option lists.

## Installation

**Main slot:** flash **Falcon4K_v3.1.1-OA.bin** with UV Studio in DFU mode, then wait for the automatic install to finish after the reboot.

**Multiboot slot:** boot on F4HWN (or a UV Studio compatible multiboot firmware), delete the old Falcon slot if there is one, install Falcon on a free slot (or the same one), turn off, boot on Falcon and wait for the automatic install to finish.

> *Falcon v3.1.1 is tested with the F4HWN 7.0.0 memory architecture. A future F4HWN update that changes it could disturb multiboot or the UV Studio online apps.*

**Falcon Apps included:**

- **FGuide**: All Falcon keys & shortcuts Guide in an App
- **FSystemInfos**: Falcon compatible System Informations App
- **FCatch**: Close Call capture of nearby transmitters, with tone, channel name, VHF/UHF modes and a saved history
- **FMessage + FMessageOption**: messaging between radios and its settings (install both)
- **FirstAid**: first aid steps (ERC 2025) and CPR metronome
- **FSurvival**: survival tips and SOS torch beacon

- **FMorse**: Morse messages by light or radio (formerly Morse)
- **FSnake**: Snake game (formerly Snake F)
- **Translations**: FirstAid and FSurvival in French, Spanish and Polish

# User guide v3.1.1

## Key reference + detailed guide

Firmware for the Quansheng **UV-K5 (V3)** and **UV-K1**, derived from Sonic and compatible with **F4HWN v7** and UV Studio.

**Notation:** Long X = press and hold X. F + X = short press on F, then on X. ◄ / ► = arrow keys. SK1 / SK2 = side keys (F1 / F2). F (#) opens the “Keys Infos” popup on the screens that offer it.

# KEY REFERENCE

## 1. Main VFO keys

| Key | Function |
|---|---|
| **0 – 9** | Enter a frequency (frequency mode) or a channel number (memory mode). Six digits = XXX.xxx MHz, confirmed automatically; MENU confirms sooner. |
| *** (during entry)** | Inserts the decimal point (e.g. 1000 * 0 = 1000.0 MHz). |
| **◄ / ►** | Frequency mode: tunes by the selected step. Memory mode: browses the channels, grouped by list. |
| **MENU** | Opens the main menu. During an entry: confirms the frequency. |
| **EXIT** | Deletes the last digit entered. With the backlight off, the first press only wakes it. |
| **Long EXIT** | Starts a new frequency entry (clears the current entry). |
| **Long 1** | Memory channel → frequency mode, on the same frequency and with the same settings. |
| **Long 2** | Switches VFO A ↔ B (DUAL, FULL WATCH and ML WATCH). |
| **Long 3** | Switches frequency mode ↔ memory mode. |
| **Long 4** | Cycles the step. In memory mode, the new step is saved to the channel. |
| **Long 5** | Switches WIDE / NARROW (FM Narrow follows SetNFM). |
| **Long 6** | Cycles the TX power (LOW 1–5, MID, HIGH). |
| **Long 7** | Turns Falcon mode on / off (SINGLE or DUAL). |
| **Long 8** | Squelch 1 → 9, looping. Level 0 and fine adjustment: Sql menu. |
| **Long 9** | Backlight ON / OFF, or on demand. |
| **Long 0** | Modulation FM → AM → USB. |
| **Long MENU** | Action set in the “M Long” menu (default: flashlight; repeat presses within 3 s to change mode). |
| **Long *** | On a memory channel: scans its list at normal speed, with the Monitor frequencies. A and B in frequency mode (DUAL): scans the range between A and B. |
| **Long F** | Keypad lock (SetLck: keys only, or keys + PTT). |

> *A long press on a digit cancels the current entry before acting. In SINGLE, Long 2 has no effect. In the status bar, SW / DW / FW / MW shows the receive mode (Single, Dual, Full Watch, ML Watch) and VX shows that VOX is on.*

## 2. F + key shortcuts

| Combination | Function |
|---|---|
| **F + MENU** | In frequency mode: creates a channel from the current settings. On a memory channel: EDIT / NEW / DEL CH. On a band slot (4067–4096): band actions (BND). |
| **F + EXIT** | Memory channels only: adds / removes its frequency from the Monitor group (“MONI ADD” / “MONI DEL”, 50 at most). No effect on an unsaved VFO frequency. |
| **F + 0** | Enters / leaves the FM radio. |
| **F + 1 to F + 4** | Makes the displayed memory channel priority channel PRICH1 to PRICH4 (“CH PRI n”), or removes it if it already is. |
| **F + 5** | Apps menu. |
| **F + 6** | Opens Sentinel (LIST / BAND / RANGE, switched with key 6). |
| **F + 7 / 8 / 9** | Opens Spectrum in LIST / BAND / RANGE mode. |
| **F + ◄ / ►** | Adjusts the VFO dBm squelch threshold (SQLSCN). The value is shown for a moment. |
| **F + SK1 / SK2** | Increases / decreases the step (saved to the channel in memory mode). |
| **F + *** | On a memory channel: fast scan (FastScan) of all eligible channels, excluding the Monitor group and the bands. In frequency mode: fast scan from the VFO frequency, with its modulation, bandwidth and step. |

> *Press F briefly: the F icon appears, then press the key you want. Priority channels are used by Full Watch mode (see Detailed guide, section 2).*

## 3. Programmable side keys and M Long

Menus **F1Shrt, F1Long, F2Shrt, F2Long** (short / long press on SK1 and SK2) and **M Long** (long press on MENU). Available actions:

| Action | Function |
|---|---|
| **NONE** | No action. |
| **FLASHLIGHT** | Flashlight (ON, SOS, BEACON modes set in the LIGHT menu). |
| **POWER** | Cycles the TX power. |
| **MONITOR** | Opens the squelch while the key is held. |
| **FM RADIO** | Enters / leaves the FM radio. |
| **LOCK KEYPAD** | Locks / unlocks the keypad. |
| **FREQ / MEM** | Switches frequency mode ↔ memory mode. |
| **MODE** | Changes the modulation. |
| **PTT TOGGLE** | Switches the PTT mode: hold to talk, or press once to transmit and once to stop. |
| **WIDE NARROW** | Switches WIDE / NARROW. |
| **ScanLst Up / Dw** | Moves to the next / previous memory list. |
| **RXMOD** | Cycles SINGLE → DUAL → FULL WATCH → ML WATCH (unavailable modes are skipped). |
| **SCAN / SCAN ALL** | Same functions as Long * and F + * in the VFO. |
| **PTT 2** | Transmits on the unselected VFO; the main PTT stays on the active VFO. |
| **HISTORY** | Opens the History from the VFO, without listening; * switches to the MESSAGES page. |
| **MESSAGES** | Opens the History directly on the MESSAGES page. |
| **FMESSAGE** | Opens the FMessage app (double beep if it is not installed). |

| Action | Function |
|---|---|
| **VOX** | Turns VOX on / off. |
| **REVERSE FQ** | The VFO receives on its TX frequency and transmits on its RX frequency. Saved with the channel; “R” on the main screen. |
| **VFO A/B** | Switches VFO A ↔ B (not offered for M Long). |

*Factory settings: F1 short = ScanLst Up, F1 long = MONITOR, F2 short = ScanLst Dw, F2 long = RXMOD, M Long = FLASHLIGHT.*

## 4. Spectrum and Sentinel

| Key | Function |
|---|---|
| **1** | Skips the current reception (temporary, nothing is deleted). |
| **2** | Backlight ON / OFF. ON: follows BLTime and lights up on RX and key presses. OFF: stays off until 2 is pressed again or the scan is reopened. |
| **3** | Speed Normal ↔ FastScan. |
| **5** | Detail screen for the current or last received frequency. |
| **6** | Cycles LIST → BAND → RANGE in the current view. |
| **7** | Saves the scanner settings. |
| **9** | Opens the History. |
| **0** | Changes the modulation for the scan session (channels and bands are not modified). |
| *** / F** | Spectrum: raises / lowers the dBm squelch (dotted line). Sentinel: raises / lowers the relative squelch (values loop). |
| **◄ / ►** | Previous / next list or band. In RANGE: shifts the whole range by its own width. |
| **SK1** | NORMAL → FREQ LOCK (stays on the last received frequency) → MONITOR (squelch open). |
| **SK2** | Blacklists the received frequency and resumes the scan. |
| **MENU** | List or band selection; in RANGE, the SCANRANGE editor. |
| **Long MENU** | Scanner parameters page (F (#) for help). |
| **PTT** | Transmits according to the PTT option: CHANNEL (VFO selected before entering) or LAST RX (last received frequency). PTT 2 has its own option. |
| **EXIT** | Back to the VFO. |

*Spectrum applies a dBm threshold on top of relative detection; Sentinel uses relative detection only. Each mode keeps its own settings and its own selections. Remember to press 7 to keep your settings.*

## 5. List and band selection

| Key | Function |
|---|---|
| ◄ / ► | Browses the entries (selected entries are shown first). |
| **4** | Adds / removes the highlighted entry from the multiple selection. |
| **MENU** | Selects only this entry and starts the scan. |
| **Long MENU** | EDIT / NEW / DEL for the list or band. |
| ***** | Select all / deselect all. |
| **EXIT** | Starts scanning the multiple selection (empty: “SELECT A LIST” or “SELECT A BAND”). |
| **F (#)** | Keys Infos. |

## 6. History

| Key | Function |
|---|---|
| **9 (in the scanner)** | Opens the History: up to 200 detected frequencies, with received code, duration, reception count and blacklist. |
| ◄ / ► | Selects an entry and listens to it with its own modulation and bandwidth; the scan pauses. |
| **MENU** | Menu: CLEAR HISTORY, CLEAR BLACKLIST, CLEAR NON BLACKLIST, SAVE HISTORY, SAVE HIST TO CH (run with MENU or 1). |
| **Long MENU** | Creates a prefilled channel (frequency, step, modulation, bandwidth, code); MENU saves it. |
| ***** | Switches to the MESSAGES page. |
| **Long *** | Starts / stops scanning the History frequencies. |
| **6** | Deletes the entry (an already saved channel is not affected). |
| **Long 1** | Sends the entry to the VFO in frequency mode, with its modulation, bandwidth and code, and closes the scanner. |
| **SK2** | Adds / removes the entry from the blacklist. |
| **SK1** | NORMAL → FREQ LOCK → MONITOR. |
| **0** | Changes the modulation. |
| **F (#)** | Keys Infos. |
| **EXIT** | During the History scan: stops it and stays on the page. Otherwise: back to the scanner, or to the VFO. |

> *Opened from Spectrum or Sentinel, the History is shown over the scanner, which keeps running: new entries keep arriving as long as none is selected. Opened from the VFO (HISTORY action), it listens to nothing: not in the background, not on the selected entry, and the LED stays off; only the History scan (Long *) listens. SAVE HISTORY keeps the History for later sessions; SAVE HIST TO CH copies every entry into free channels.*

## 7. MESSAGES

| Key | Function |
|---|---|
| *** (in the History)** | Opens the MESSAGES page; * returns to the History. The scan and the receiver pause on this page. |
| **◄ / ►** | Selects a message. |
| **F then ◄ / ►** | Jumps to the newest or oldest message. |
| **MENU** | Opens the selected message in full, under “FROM” and the sender’s callsign. |
| **6** | Deletes the message: DELETE? popup, MENU confirms, EXIT cancels. Also works on an open message. |
| **Long 6** | Clears all the messages: CLEAR ALL? popup, MENU confirms. |
| **EXIT** | Closes the open message; on the list: leaves (back to the scanner or the VFO). |

*Messages received by the FMessage app are saved automatically in external flash. The page lists the 32 newest, newest first: number, sender’s callsign and the start of the text. A dotted line marks each new FMessage session. The F1/F2/M long MESSAGES shortcut opens this page directly from the VFO. No scan, no listening and no LED on this page.*

## 8. Falcon mode

| Action | How to |
|---|---|
| **Turn on** | Long 7 in SINGLE or DUAL: 3 green flashes. The backlight and flashlight turn off. |
| **Turn off** | Short EXIT or Long 7: normal controls come back. |
| **Switch A / B** | In DUAL, Long 2: 1 red flash = VFO A, 2 slower red flashes = VFO B. |
| **Active keys** | PTT and Long 2 (in DUAL) only; all other keys are blocked. |
| **Indications** | Reception: brief green flash. Transmission: red LED, limited to 1 s. |

## 9. FM radio

| Key | Function |
|---|---|
| **F + 0 / EXIT** | Enters the FM radio / returns to the VFO. |
| **◄ / ►** | Auto mode: next station. Manual mode: 0.1 MHz steps, wrapping at the band limits. |
| ***** | Switches auto seek ↔ manual tuning. |
| **1 – 9** | Recalls a favorite (no effect if empty). |
| **Long 1 – 9** | Saves the station to that favorite (replaces the previous one). |
| **MENU** | Shows / hides the station name. |
| **0** | Turns background VFO reception on / off; the FM radio keeps playing. |
| **PTT** | Transmits on the radio VFO (never on the FM station), then FM listening resumes. |

*While listening to FM, actions that would change the VFO radio settings are blocked; the flashlight and other compatible actions remain available.*

## 10. Editors (channel, list, band)

| Key | Function |
|---|---|
| ◄ / ► | Previous / next field. |
| **1 / 3** | Previous / next value. On NAME: moves the cursor. |
| **4 / 6 (NAME)** | Scrolls through letters (hold to repeat). |
| **7 / 9 (NAME)** | Scrolls through space, digits and symbols. |
| **Digits (frequency)** | Enters the frequency or offset, up to 5 decimal places. |
| *** (frequency)** | Inserts the decimal point. Enter all 5 decimals to avoid rounding to the step. |
| **MENU** | Checks, saves and exits. On error, the editor stays open for correction. |
| **EXIT** | Deletes the pending digit; otherwise exits without saving. |
| **Long EXIT** | Exits without saving. |
| **F (#)** | Keys Infos (not in SCANRANGE). |
| **PTT** | Exits without saving and transmits. |
| **Long 4 / Long MENU** | In a list's channel list (EDIT LST): puts the channel in Monitor / opens it to manage it. |

# DETAILED GUIDE

## 1. Single and dual VFO

Each VFO works in frequency mode (free frequency) or memory mode. **Long 3** switches between the two, keeping the remembered state; **Long 1** copies the displayed channel exactly into frequency mode, with its settings. In DUAL, FULL WATCH and ML WATCH, **Long 2** changes the active VFO (A or B).

Direct six-digit entry works in both layouts. 5, 50 and 500 are confirmed as 5, 50 and 500 MHz; * inserts the decimal point for four-digit MHz frequencies (1000 * 0). In memory mode, the arrows browse the channels grouped by list, without renumbering the memory.

## 2. Full Watch and Multilist Watch

Set **RxMode = FULL WATCH**: the radio watches A, B and up to four priority channels (PRICH1 to PRICH4), six channels at most. **ML WATCH** scans the memory lists of A and B in the background.

| Item | Function |
|---|---|
| **PRICH1 to PRICH4** | Ordinary memory channels (not bands) or NONE. Set in the menu or with F + 1 to F + 4 from the VFO. |
| **Display** | In Full Watch, priority channels appear as numbered bubbles in the middle of the screen. ML WATCH shows no bubbles. |
| **Reception** | A watched channel that receives moves to the active VFO; the one it replaces stays in the watch cycle. |
| **Long 2** | Changes the selected VFO A / B, without clearing the priorities. |
| **Cycle** | Channels are checked one after another, not at the same time: there is only one receiver. |
| **Restrictions** | Falcon mode is only available in SINGLE and DUAL. |

### Multilist Watch rules

- RXMOD cycles Single → Dual → Full Watch → ML Watch → Single; unavailable modes are skipped.
- With two different lists: list A → Monitor → PRICH1 → … → PRICH4 → list B → Monitor → PRICH1 → … → PRICH4, then again. Each list keeps its position.
- If A and B are in the same list, it is scanned only once (through A).
- A memory without a list selects the unlisted channels. Band slots and internal Monitor copies are excluded.
- A VFO in frequency mode contributes no list; the other one can still be scanned. With no list at all, the radio returns to Dual.
- Valid priorities that are not in the scanned lists are checked between the lists; a duplicate is checked only once.
- At rest, A and B show their original frequency. A reception shows the received channel on its VFO, then the display returns when the reception ends.
- PTT always uses the selected VFO and its transmit settings, never the temporarily received channel.
- Changing VFO does not interrupt reception on the other one; operating the receiving VFO, or EXIT, can interrupt it.
- SQLSCN sets the scan opening threshold; SQLHYS adds a 0 to 8 dBm holding margin (opens at the threshold, closes below threshold − SQLHYS).
- Manual scan, Spectrum, Sentinel, FM radio, menus and apps take over; ML Watch resumes when you return to the VFO screen.
- Falcon mode is turned off when entering ML Watch. The scan runs continuously in FastScan, without the Dual Watch power-saving cycle.

## 3. Falcon mode

Discreet, battery-saving mode for simple RX / TX exchanges: backlight off, almost all keys blocked, LED indications only. It uses the selected SINGLE or DUAL reception.

## 4. Spectrum and Sentinel

| Mode / view | Description |
|---|---|
| **LIST** | Scans the channels of one or more lists. Each channel brings its modulation and step; the listening bandwidth comes from ListenBW. |
| **BAND** | Scans one or more saved bands: limits, modulation and step. |
| **RANGE** | Scans a temporary range; MENU opens SCANRANGE (FSTART, FSTOP, STEP, MOD, BANDW). |
| **Spectrum** | Full-width activity graph, with a footer (selected lists or bands, or the range limits). |
| **Sentinel** | Compact display: active frequency, progress or S-meter, last reception, counter. The timer counts the time since the last reception. |
| **Normal / FastScan** | Key 3 changes the speed. FastScan measures the signal more briefly; actual speed depends on the scan and the signals. |

### Scans started from the VFO

Long * scans the channel's list at normal speed; F + * scans all eligible channels in FastScan; in frequency mode, F + * starts from the VFO frequency with its settings. The VFO display is kept: 1 skips the reception, SK2 blacklists the frequency, EXIT or another key stops the scan. With A and B in frequency mode, Long * scans the range between them; the blacklist of this scan is temporary and separate from the History blacklist.

## 5. History and blacklists

The History keeps up to 200 scanner detections: frequency, received code, blacklist, duration and reception count. VFO receptions in SINGLE, DUAL or FULL WATCH are not added to it. SK2 toggles the blacklist, Long MENU creates a prefilled channel, Long 1 sends the entry to the VFO.

## 6. Messages

The MESSAGES page keeps the messages received by FMessage (see the Key reference, section 7), in the external flash area that F4HWN v7 leaves free. Install FMessage and FMessageOption to send and receive messages; the FMESSAGE shortcut opens the app from the VFO.

## 7. VOX

VOX menu: OFF or levels 1 to 10. VOX works in FM and shows "VX" in the status bar; a side key can toggle it with the VOX action. Sensitivity also depends on the microphone gain (Mic menu): if key clicks trigger short transmissions, lower Mic or choose another VOX level.

## 8. Channel, list and band editors

### Create or edit a channel

From a VFO in frequency mode, F + MENU creates a channel from the current settings. From a memory, F + MENU offers EDIT CH, NEW CH and DEL CH: choose with the arrows and confirm with MENU (DEL asks for a second confirmation). A new channel assigned to a list is placed after that list's channels when space is left.

| Field | Function |
|---|---|
| **LIST and NAME** | List membership (or BAND / MONITOR destination where offered) and channel name (7 characters; lists: 6, bands: 8). |
| **STEP and FREQ** | Step and RX frequency. An incomplete entry may be rounded to the step; 5 decimals keep the exact value. |
| **MOD and BAND** | FM, AM or USB, then WIDE / NARROW bandwidth. |
| **POWER** | LOW 1–5, MID or HIGH (actual output depends on calibration and frequency). |
| **OFFSET** | TX offset and direction OFF, + or −. |
| **REVERSE** | OFF or ON: receives on the TX frequency and transmits on the RX frequency (as the REVERSE FQ shortcut); "R" on the main screen. |

| Field | Function |
|---|---|
| **CTCSS RX / TX** | OFF or a tone, separately for receive and transmit; replaces a DCS code in the same direction. |
| **DCS RX / TX** | OFF or a normal / inverted DCS code, separately for RX and TX; replaces a CTCSS tone in the same direction. |

### Create or edit a list

In Spectrum or Sentinel LIST mode, MENU opens the selection; Long MENU offers EDIT LST, NEW LST and DEL LST. NEW asks for a free number from 1 to 25 and a name of up to 6 characters. Changing a list's number moves its channels to it without changing their frequencies. DEL LST deletes the list and its channels, after confirmation. In EDIT LST, Long 4 puts a channel in Monitor or removes it, Long MENU opens it to manage it.

### Create or edit a band

In Spectrum or Sentinel BAND mode, Long MENU in the selection offers EDIT BND, NEW BND and DEL BND. The same actions open with F + MENU on memories 4067 to 4096, which hold the 30 band definitions.

| Field | Function |
|---|---|
| **Slot and NAME** | Band slot and optional name (8 characters). A new band needs a free slot. |
| **STEP** | Sweep step. |
| **FSTART / FSTOP** | Valid RX limits, FSTOP strictly greater than FSTART. |
| **MOD / BANDW** | Modulation and bandwidth used to scan the band. |
| **Save / delete** | MENU saves. DEL BND asks for confirmation and does not erase the channels inside the band. |

### Temporary range (SCANRANGE)

In RANGE, MENU opens SCANRANGE: FSTART, FSTOP, STEP, MOD and BANDW. Arrows change the field, 1 / 3 change the values, MENU applies the range. EXIT first deletes the pending digit, then cancels.

### Monitor frequencies and priority channels

On a memory channel, F + EXIT makes a Monitor copy in a free slot, keeping the original channel's list; a second F + EXIT removes it. The group holds 50 frequencies at most ("MONI FULL"), depending on free space; band slots cannot be added. Turn on MONITOR CH in the scanner parameters to include it in scans. F + 1 to F + 4 assign the displayed channel to PRICH1 to PRICH4 the same way.

## 9. FM radio

Enter with F + 0 or a side key set to FM RADIO. Nine favorites, separate from the radio channels, auto seek or 0.1 MHz steps, station name and reception measurements. VFO reception can stay active in the background (key 0).

## 10. Falcon-specific menu entries

| Menu | Function |
|---|---|
| **RxMode** | SINGLE, DUAL, FULL WATCH or ML WATCH. |
| **PRICH1 to PRICH4** | Priority channels for Full Watch and ML Watch. |
| **F1Shrt … M Long** | Side-key and MENU long-press actions (Key reference, section 3). |
| **POWER** | Power LOW 1–5, MID or HIGH. |
| **Sql / SQLSCN / SQLHYS** | Squelch in levels (fixed VFO); dBm opening threshold for the VFO and its scans; 0–8 dBm holding margin. |
| **VOX** | OFF or levels 1 to 10. |
| **Roger / AUROG** | End-of-transmission beep (OFF, STARWARS, R2D2, FALCON, ROGGER, OURO); automatic Roger transmission every 10 s, 30 s, 2, 5, 10 or 30 min. |
| **Tone** | OFF or 1750 Hz: SK2 sends the tone while transmitting. |
| **LIGHT** | Flashlight mode: ON, SOS, BEACON, OFF. |
| **SetCtr / SetVol / SetMet** | Screen contrast; receive volume; S-meter style. |
| **BatTyp** | Battery model and capacity (UV-K1 / UV-K5). |
| **ListenBW FM / AM** | In the scanner parameters: FM listening bandwidth (25k / 12.5k / 6.25k) and AM (Wide / Narrow). |

## Appendix: differences from Sonic

| Topic | In Falcon |
|---|---|
| **Ultra Watch** | Becomes Sentinel, with LIST / BAND / RANGE and a compact display. |
| **Scan access** | F + 7 / 8 / 9: Spectrum LIST / BAND / RANGE; F + 6: Sentinel. The SPECTRUM / ULTRA WATCH / SMOOTH SPECTRUM cycle and the key-8 view switch are removed. |
| **History** | Compact rows, Keys Infos; Long * scans, MENU opens the actions, Long MENU creates a channel, 6 deletes, SK2 handles the blacklist. Close Call and the old shortcuts are removed. |
| **Lists and bands** | 25 lists (6-character names) and 30 bands (8 characters) in memories 4067 to 4096. |
| **VFO entry** | Direct six-digit entry in single and dual VFO; Long 1 copies the memory, Long 3 switches frequency / memory. |
| **Navigation** | Memories grouped by list, without renumbering the memory. |
| **RF filters** | F4HWN FM and AM Narrow profiles for the UV-K1; scanner's own AM listening filter. |
| **Display** | CTCSS / DCS codes shown instead of AM / FM when useful; F4HWN-style screen refresh. |
| **Startup** | The radio restarts on the VFO; the splash screen shows the battery and the CPU temperature. |
| **Additions** | On-radio editors, Full Watch, ML Watch, Falcon mode, Monitor group, ListenBW AM, SetVol, AUROG, POWER, SetMet, LIGHT and Tone menus. |

*Guide written for Falcon 4K v3.1.1 from the Falcon USER GUIDE and the firmware code. Use it as a quick reference: always check the behavior on your radio.*